SMART DUSTBIN

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➢ Introduction

- A Smart Dustbin is a simple electronic dustbin that can open its lid automatically.
- It uses an ultrasonic sensor to detect an object or hand.
- A servo motor is used to move the dustbin lid.
- LED lights show the working/status of the system.
- The components are connected using jumper wires and a breadboard.
- It helps reduce direct contact with the dustbin.

➢ Objective of the Project

- ❑ The main objectives of our Smart Dustbin are:
- ❑ To make a basic automatic dustbin.
- ❑ To open the lid without touching it.
- ❑ To detect a hand/object using an ultrasonic sensor.
- ❑ To control the lid using a servo motor.
- ❑ To provide a simple and low-cost electronic project.
- ❑ To improve basic cleanliness and hygiene.

➢ Components Used

❑ Component	❑ Purpose
• Ultrasonic Sensor	• Detects the hand/object
• Servo Motor	• Opens and closes the lid
• Breadboard	• Used for circuit connections
• Jumper Wires	• Connects the components
• LED Lights	• Shows the system status

➢ Ultrasonic Sensor

- ❑ It is used to detect the distance of an object.
- ❑ It sends ultrasonic waves and receives the reflected waves.
- ❑ When a hand comes near the dustbin, the sensor detects it.
- ❑ The detection signal is used to operate the servo motor.

Main Function:

👉 **Detects the hand/object near the dustbin.**

➢ Servo Motor

- A servo motor is used to control the movement of the dustbin lid.
- When the ultrasonic sensor detects a hand, the servo motor rotates.
- This rotation opens the lid.
- After a short time, the servo returns to its original position and closes the lid.

Main Function:

👉 **Opens and closes the dustbin lid automatically.**

➢ Breadboard, Jumper Wires & LED

- **Breadboard**

❑ Used to make the circuit easily.
❑ No soldering is required.

- **Jumper Wires**

❑ Used to connect the different components.
❑ They carry power and signals between components.

- **LED Lights**

❑ Indicate that the system is working.
❑ They provide a simple visual indication.

➢ Working of Basic Smart Dustbin

- **Step 1:** The system is switched ON.
- **Step 2:** The ultrasonic sensor checks for an object/hand.
- **Step 3:** When a hand comes near the dustbin, the sensor detects it.
- **Step 4:** The control signal activates the servo motor.
- **Step 5:** The servo motor opens the dustbin lid.
- **Step 6:** After a short delay, the servo motor closes the lid.
- **Step 7:** The LED indicates the system's working status.

➢ Advantages

- ❑ Simple and easy to use.
- ❑ Touch-free operation.
- ❑ Low-cost project.
- ❑ Easy to build.
- ❑ Reduces direct contact with the dustbin.
- ❑ Helps maintain basic hygiene.
- ❑ Components are easily available.
- ❑ Suitable for school/college projects.

➢ Conclusion

- ❑ The Basic Smart Dustbin is a simple automation project.
- ❑ The ultrasonic sensor detects a hand or object.
- ❑ The servo motor opens and closes the lid.
- ❑ LED lights indicate the system status.
- ❑ Breadboard and jumper wires are used for circuit connections.
- ❑ This project demonstrates how simple electronic components can be used to create an automatic system.


thank
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